Graph-Attention Network with Adversarial Domain Alignment for Robust Cross-Domain Facial Expression Recognition

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1. Introduction

Cross-domain facial expression recognition (CD-FER) remains challenging due to severe domain shift between training and deployment data. We propose **GAT-ADA** (Graph-Attention Network with Adversarial Domain Alignment), a hybrid framework coupling ResNet-50 with a **batch-level Graph Attention Network (GAT)** to model inter-sample relations under shift. Each mini-batch is cast as a sparse ring graph where attention aggregates cross-sample cues for adaptation. To align distributions, GAT-ADA combines **adversarial learning (GRL)** with **statistical alignment (CORAL + MMD)**. Evaluated on RAF-DB$\rightarrow${CK+, JAFFE, SFEW 2.0, FER2013, ExpW}, GAT-ADA achieves **74.39% mean accuracy** and **98.0% on FER2013** (≈36-point improvement over best baseline).

2. Motivation & Aim

Problem: Current CD-FER methods face critical limitations:

- CNNs overlook inter-sample relational structure across domains
- Existing methods rely only on marginal distribution alignment
- Typical CD-FER accuracy: $<$60% (vs. $>$90% within-domain)

Contributions

⇒ **Hybrid Graph-Based Modeling**: Batch-level GAT treating each image as a node to capture inter-sample relationships across domains

⇒ **Multi-Component Domain Adaptation**: CORAL + MMD + GRL for comprehensive marginal and statistical distribution alignment

⇒ **Comprehensive Evaluation**: Systematic assessment across 6 benchmark datasets (RAF-DB, CK+, JAFFE, SFEW 2.0, FER2013, ExpW)

4. Methodology: GAT-ADA Architecture

Three Main Components

1. Feature Extraction (ResNet-50): Extract 2048-D feature maps → GAP → 512-D embeddings

2. Batch-Level Graph Attention Network: Each image = graph node; Ring topology with $\mathcal{O}(n)$ edges and Multi-head attention aggregates cross-sample information

3. Multi-Component Domain Alignment:Adversarial: GRL with domain classifier and Statistical: CORAL (covariance) + MMD (mean matching)

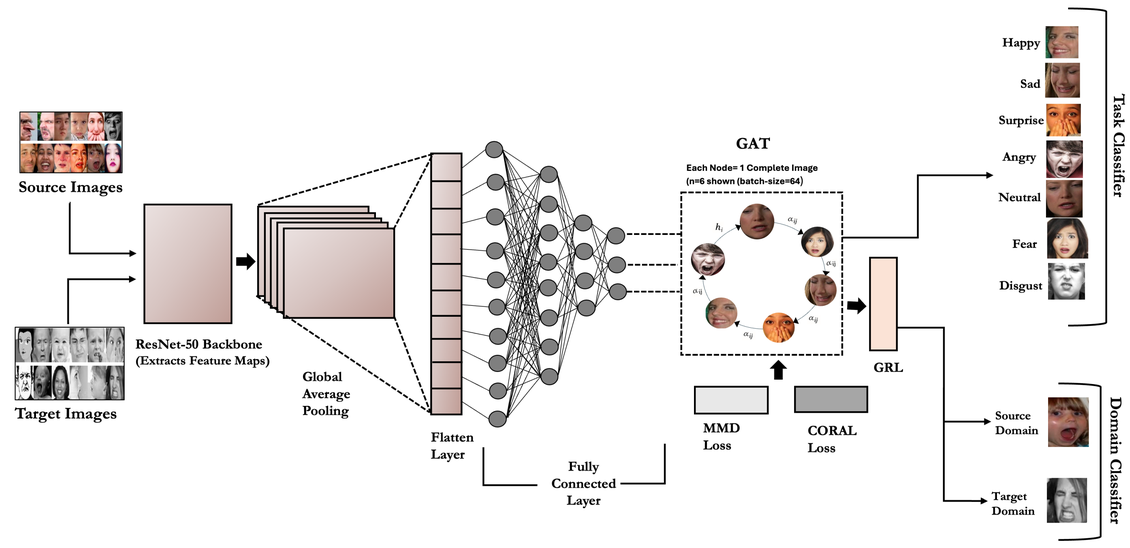


Figure 1. Pipeline of the proposed GAT-ADA framework showing the three main components: ResNet-50 feature extractor, batch-level GAT for relational modeling, and multi-component domain adaptation.

5. Experimental Results

Component Analysis (RAF-DB to FER2013):

Table 1. Impact of different components on performance

Configuration	Accuracy	AUC
ResNet-50 + GAT + GRL	**98.04%**	**0.9804**
ResNet-50 + GAT (no GRL)	64.51%	0.7564
ResNet-50 + GCN + GRL	61.04%	0.7141
ConvNeXt + GAT + GRL	96.74%	0.9799
ResNet-18 + GAT + GRL	98.68%	0.9919

Critical Insights:

GRL Impact: Removing GRL drops accuracy by 33.53% (98.04→64.51)
GAT vs GCN: Attention mechanism crucial for CD-FER (98.04% vs 61.04%)

3. Datasets

Source: RAF-DB (labeled)
Targets:CK+, JAFFE, SFEW 2.0, FER2013, ExpW (unlabeled)



Figure 2. Facial expression examples from six datasets: CK+, JAFFE, SFEW2.0, FER2013, ExpW, and RAF-DB showing 7 emotions

5.1 Benchmark Comparison

Table 2. Comparison with state-of-the-art CD-FER methods (ResNet-50 backbone)

Method	CK+	JAFFE	SFEW2.0	FER2013	Mean
AGRA (2022)	85.27	61.50	56.43	58.95	66.13
AGLRLS (2024)	87.60	61.97	58.28	60.68	68.30
FER-DAS (2025)	89.25	69.93	57.41	61.80	69.60
GAT-ADA (Ours)	**88.04**	54.06	**58.14**	**98.04**	**74.39**

5.2 Computational Performance

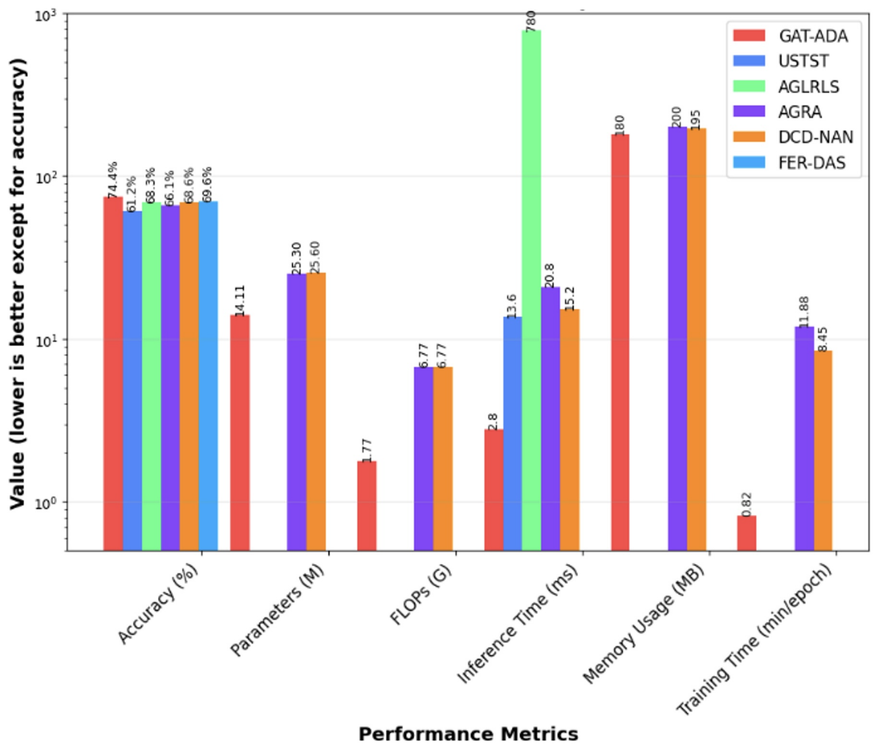


Figure 3. Comparing GAT-ADA with AGRA, DCD-DAN, USTST, AGLRLS shows the strongest balance, high mean accuracy with markedly lower FLOPs, latency, and memory supporting real-time, resource-constrained deployment.

6. Opportunities for Further Research

- Extend to multimodal emotion recognition (audio-visual)
- Improve robustness on small datasets (JAFFE, SFEW 2.0)
- Graph sparsification for enhanced efficiency
- Explore transferability to other domain adaptation problems